AAC ASSIGNMENT 8.3

V.Harinadh

2303A52302

BATCH-45

Task 1: Email Validation using TDD

Scenario

You are developing a user registration system that requires reliable email input validation.

Requirements

• Must contain @ and . characters

• Must not start or end with special characters

• Should not allow multiple @ symbols

• AI should generate test cases covering valid and invalid email formats

• Implement is_valid_email(email) to pass all AI-generated test cases

Expected Output

• Python function for email validation

• All AI-generated test cases pass successfully

• Invalid email formats are correctly rejected

• Valid email formats return True

Prompt:

You are a senior Python developer following Test-Driven Development (TDD).

Task:

Implement email validation using TDD.

Requirements:

1. Email must contain exactly one '@' symbol.

2. Email must contain at least one '.' after the '@'.

3. Email must not start or end with special characters like '.', '@', '-', '_'.

4. Email must not contain multiple '@' symbols.

5. Email should have characters before and after '@'.

6. Domain part must contain at least one '.'.

7. No spaces allowed in the email.

Instructions:

1. First, generate comprehensive test cases (both valid and invalid emails).

2. Cover edge cases.

3. Then implement the function:

```
def is_valid_email(email: str) -> bool
```

4. Follow TDD approach:

- Write test cases first
- Then implement the function
- Ensure all tests pass

5. Print results of all test cases clearly showing:

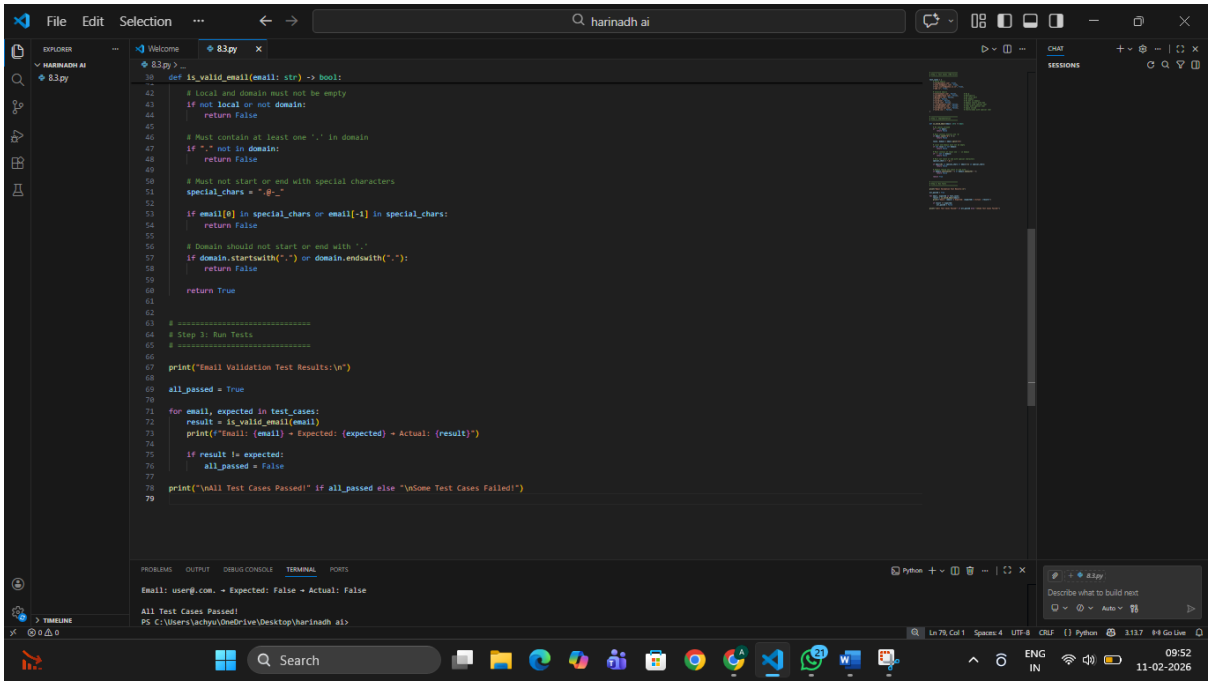
Email: <email> → Expected: <True/False> → Actual: <True/False>

6. Use only Python standard library (no external libraries).

Output format:

- Test cases list
- Function implementation
- Test execution results

Output:



Task 2: Grade Assignment using Loops

Scenario

You are building an automated grading system for an online examination platform.

Requirements

• AI should generate test cases for assign_grade(score) where:

– 90–100 → A

– 80–89 → B

– 70–79 → C

– 60–69 → D

– Below 60 → F

• Include boundary values (60, 70, 80, 90)

• Include invalid inputs such as -5, 105, "eighty"

• Implement the function using a test-driven approach

Expected Output

• Grade assignment function implemented in Python

• Boundary values handled correctly

• Invalid inputs handled gracefully

• All AI-generated test cases pass

Prompt:

You are a senior Python developer following Test-Driven Development (TDD).

Task:

Build a grade assignment system.

Function to implement:

```
def assign_grade(score)
```

Grading Rules:

90–100 → A

80–89 → B

70–79 → C

60–69 → D

Below 60 → F

Requirements:

1. Generate comprehensive test cases first.

2. Include boundary values: 60, 70, 80, 90.

3. Include edge values: 59, 69, 79, 89, 100.

4. Include invalid inputs:

   - Negative numbers (-5)

   - Numbers greater than 100 (105)

   - Non-numeric input ("eighty")

5. Handle invalid inputs gracefully (return "Invalid Input").

6. Implement the function using loops where appropriate.

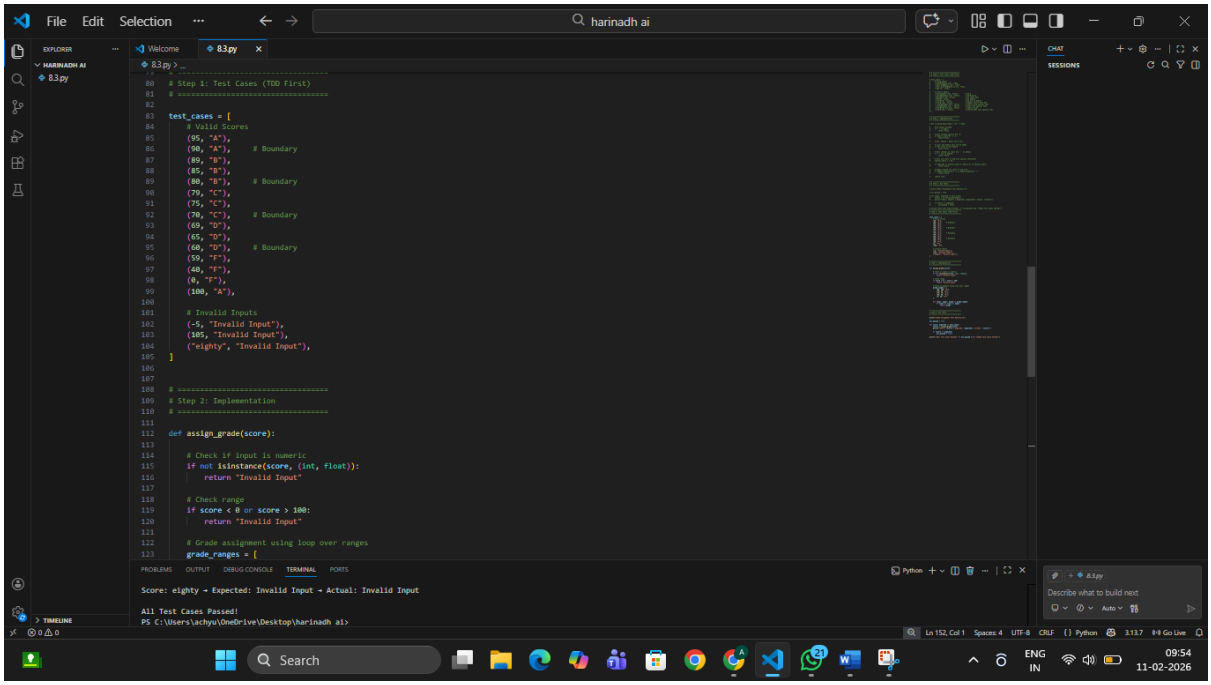
7. Print test results clearly in the format:

   Score: <score> → Expected: <grade> → Actual: <grade>

8. Ensure all test cases pass.

9. Use only Python standard library.

Output:



Task 3: Sentence Palindrome Checker

Scenario

You are developing a text-processing utility to analyze sentences.

Requirements

• AI should generate test cases for is_sentence_palindrome(sentence)

• Ignore case, spaces, and punctuation

• Test both palindromic and non-palindromic sentences

• Example:

– "A man a plan a canal Panama" → True

Expected Output

• Function correctly identifies sentence palindromes

• Case and punctuation are ignored

• Returns True or False accurately

• All AI-generated test cases pass

Prompt:

You are a senior Python developer following Test-Driven Development (TDD).

Task:

Implement a function:

```
def is_sentence_palindrome(sentence: str) -> bool
```

Requirements:

1. Ignore:
   - Case (uppercase/lowercase)
   - Spaces
   - Punctuation
2. Only consider alphanumeric characters.
3. Generate comprehensive test cases first.
4. Include:
   - Simple palindromes
   - Sentence palindromes
   - Non-palindromes

- Empty string

- Only punctuation

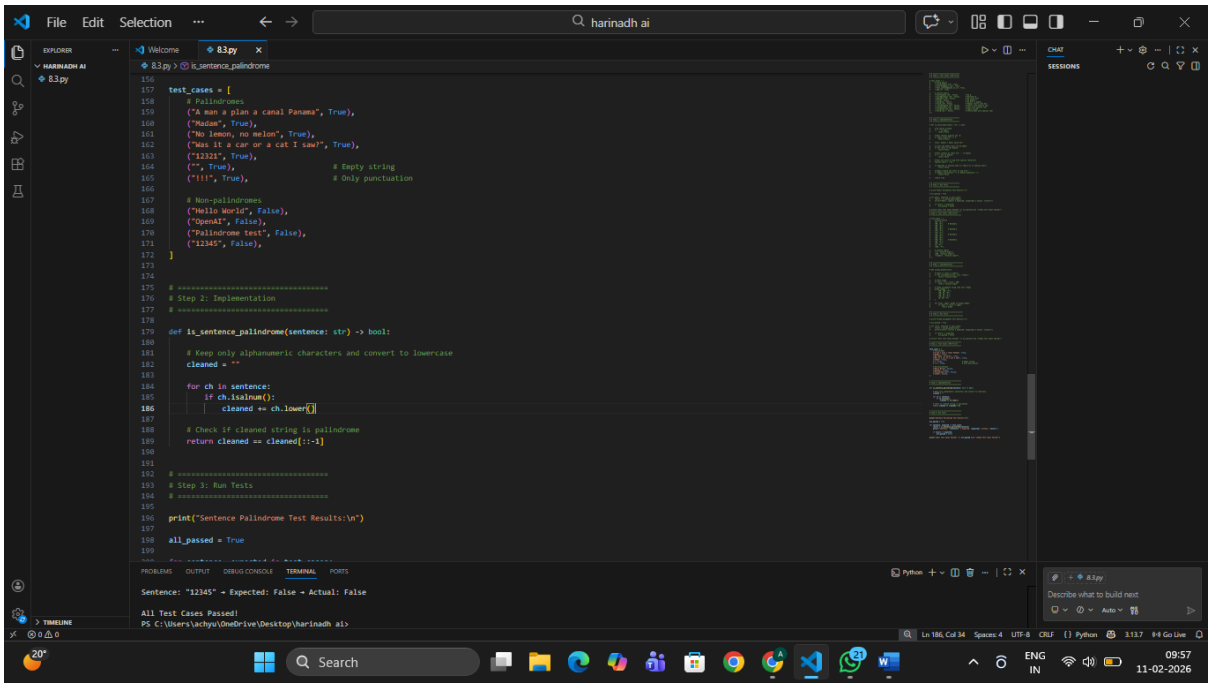
5. Print results in format:

Sentence: "<sentence>" → Expected: <True/False> → Actual: <True/False>

6. Use only Python standard library.

7. Ensure all test cases pass.

Output:



Task 4: ShoppingCart Class

Scenario

You are designing a basic shopping cart module for an e-commerce application.

Requirements

• AI should generate test cases for the ShoppingCart class

• Class must include the following methods:

– add_item(name, price)

– remove_item(name)

– total_cost()

• Validate correct addition, removal, and cost calculation

• Handle empty cart scenarios

Expected Output

• Fully implemented ShoppingCart class

• All methods pass AI-generated test cases

• Total cost is calculated accurately

• Items are added and removed correctly

Prompt:

You are a senior Python developer following Test-Driven Development (TDD).

Task:

Design a ShoppingCart class.

Class Requirements:

1. add_item(name, price)

2. remove_item(name)

3. total_cost()

Rules:

- Item name must be string.

- Price must be positive number.

- Removing a non-existing item should be handled gracefully.

- total_cost() should return 0 for empty cart.

- Support multiple items.

Instructions:

1. First generate comprehensive test cases:

   - Add single item

   - Add multiple items

   - Remove item

   - Remove non-existing item

   - Empty cart total

   - Invalid price

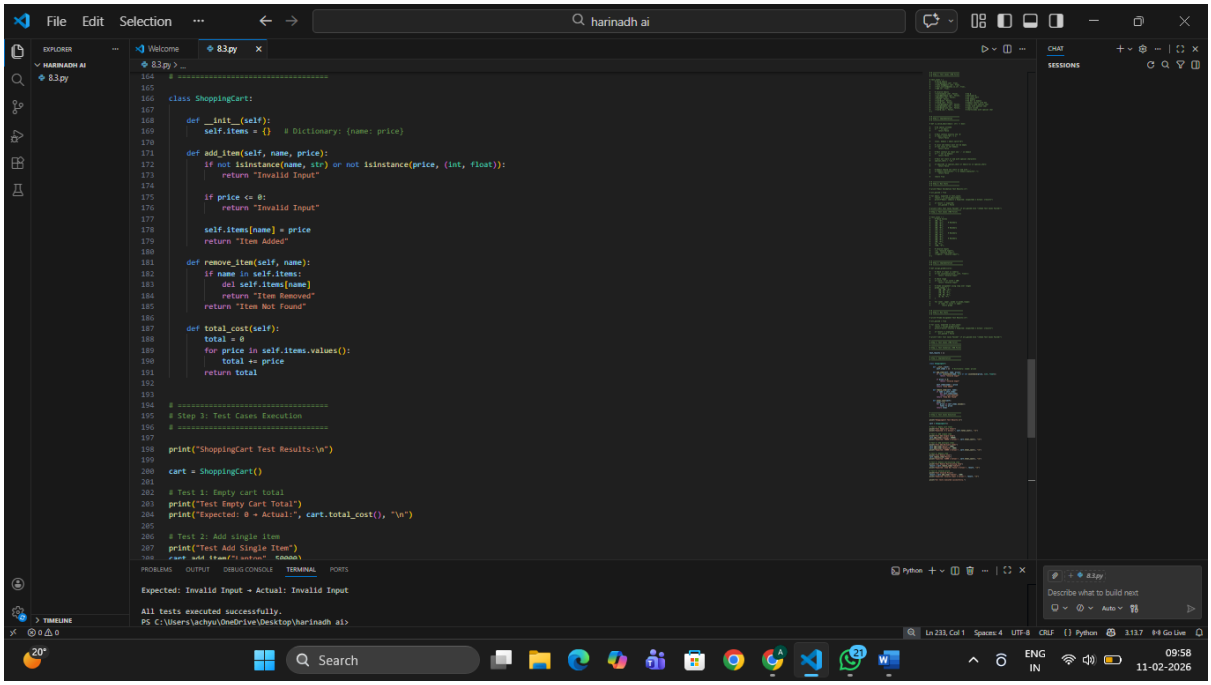
2. Implement the ShoppingCart class.

3. Print test results clearly.

4. Ensure all test cases pass.

5. Use only Python standard library.

Output:



Task 5: Date Format Conversion

Scenario

You are creating a utility function to convert date formats for reports.

Requirements

• AI should generate test cases for convert_date_format(date_str)

• Input format must be "YYYY-MM-DD"

• Output format must be "DD-MM-YYYY"

• Example:

– "2023-10-15" → "15-10-2023"

Expected Output

• Date conversion function implemented in Python

• Correct format conversion for all valid inputs

• All AI-generated test cases pass successfully

Prompt:

You are a senior Python developer following Test-Driven Development (TDD).

Task:

Implement a function:

```
def convert_date_format(date_str: str) -> str
```

Requirements:

1. Input format must be exactly "YYYY-MM-DD".
2. Output format must be "DD-MM-YYYY".
3. Generate comprehensive test cases first.
4. Include:
   - Valid dates
   - Edge dates like "2023-01-01"
   - Invalid formats like:
     "15-10-2023"
     "2023/10/15"
     "2023-13-01"
     "2023-00-10"
     "abcd-ef-gh"
5. Handle invalid inputs gracefully (return "Invalid Date Format").
6. Print results in format:
   Input: <date> → Expected: <output> → Actual: <output>
7. Use only Python standard library.
8. Ensure all test cases pass.

Output:

File Edit Selection ... harinadh ai

EXPLORER
HARINADH AI
8.3.py

Welcome | 8.3.py

8.3.py > ...

```python
#     if result != expected:
#         all_passed = False

# print("\nAll Test Cases Passed!" if all_passed else "\nSome Test Cases Failed!")

# ==================================
# Step 1: Test Cases (TDD First)
# ==================================
# ==================================
# Step 1: Test Scenarios (TDD First)
# ==================================

test_results = []

# ==================================
# Step 2: Implementation
# ==================================

class ShoppingCart:

    def __init__(self):
        self.items = {}   # Dictionary: {name: price}

    def add_item(self, name, price):
        if not isinstance(name, str) or not isinstance(price, (int, float)):
            return "Invalid Input"

        if price <= 0:
            return "Invalid Input"

        self.items[name] = price
        return "Item Added"

    def remove_item(self, name):
        if name in self.items:
            del self.items[name]
            return "Item Removed"
        return "Item Not Found"

    def total_cost(self):
        total = 0
        for price in self.items.values():
            total += price
        return total
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Expected: Invalid Input → Actual: Invalid Input

All tests executed successfully.
PS C:\Users\achyu\OneDrive\Desktop\harinadh ai>
```

CHAT
SESSIONS
Describe what to build next

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